

Parameter-Efficient pretrained-CT-to-MRI Transfer for Rectal Cancer Segmentation: Performance Calibration Trade-offs

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Medical Image AI in Radiation Therapy



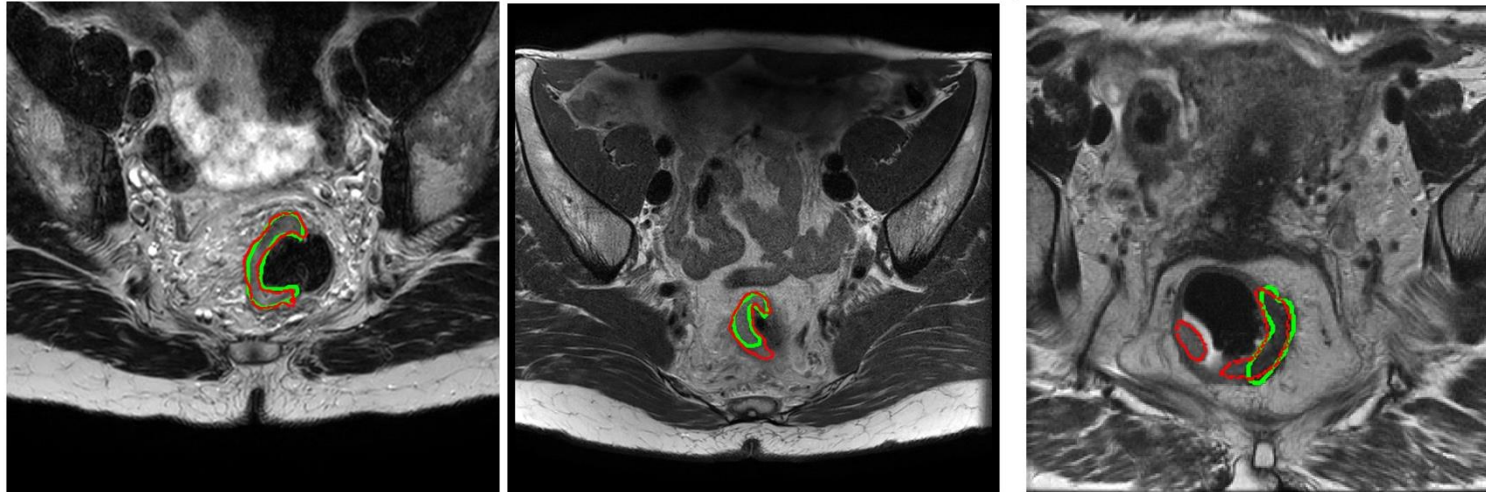
Practical challenges of MR based rectal cancer segmentation

- Images and tumors vary substantially across treatment
- Carefully curated expert contours are limited

Pre-RT

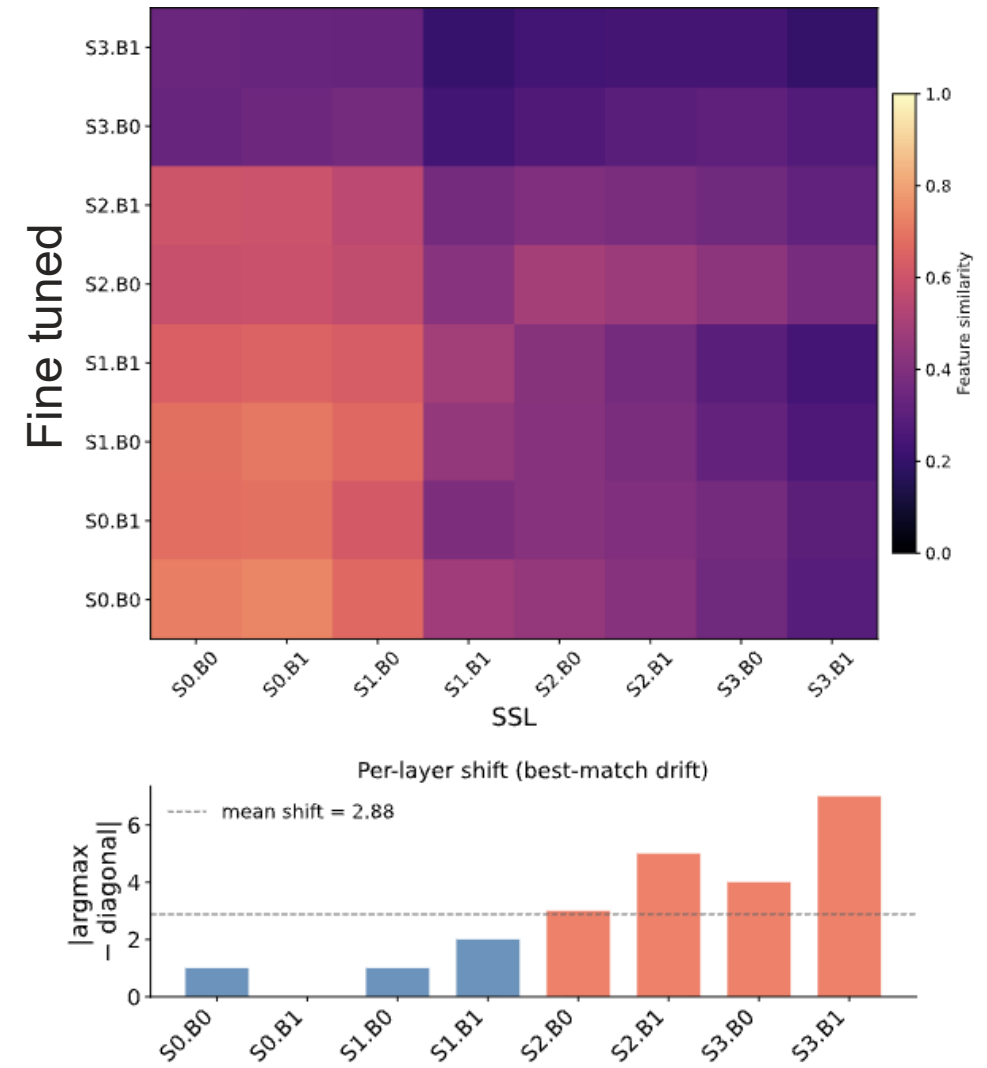
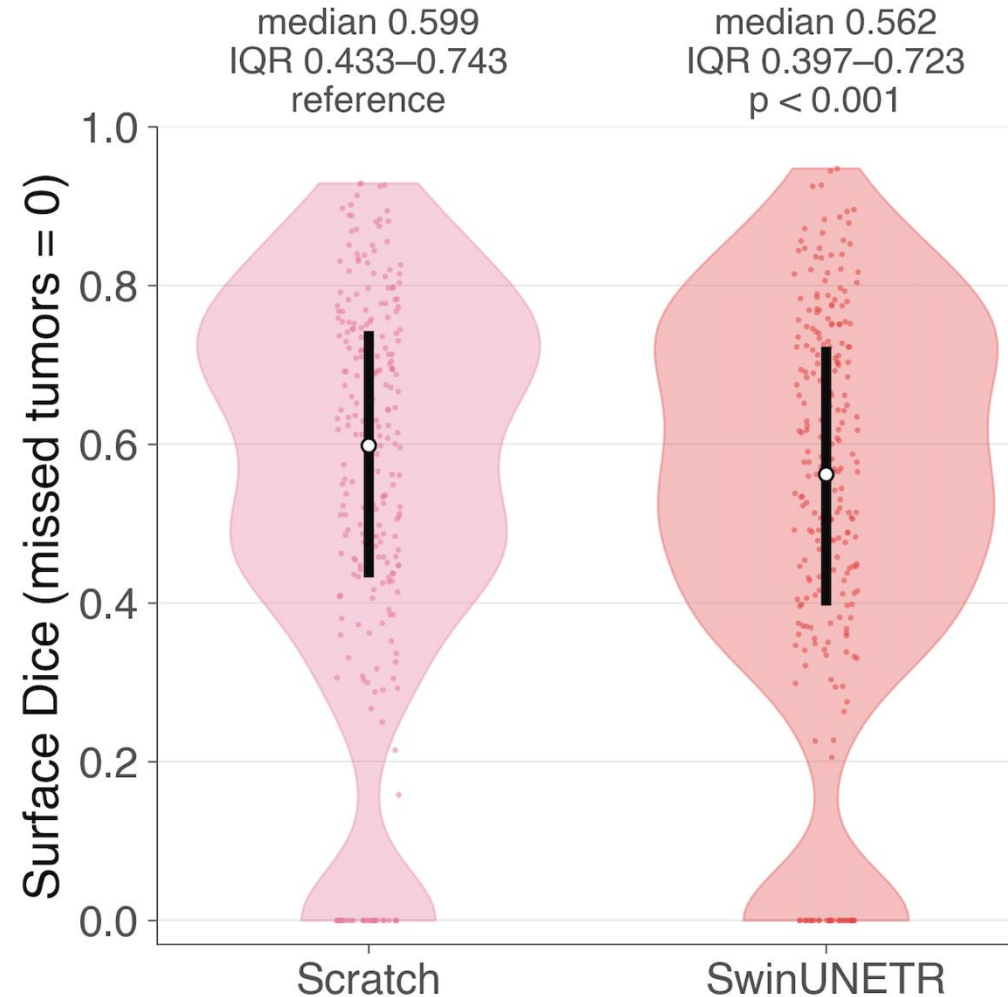


Mid or
Post RT



Practical challenges of MR based segmentation

Domain gap diminishes returns of pretrained models

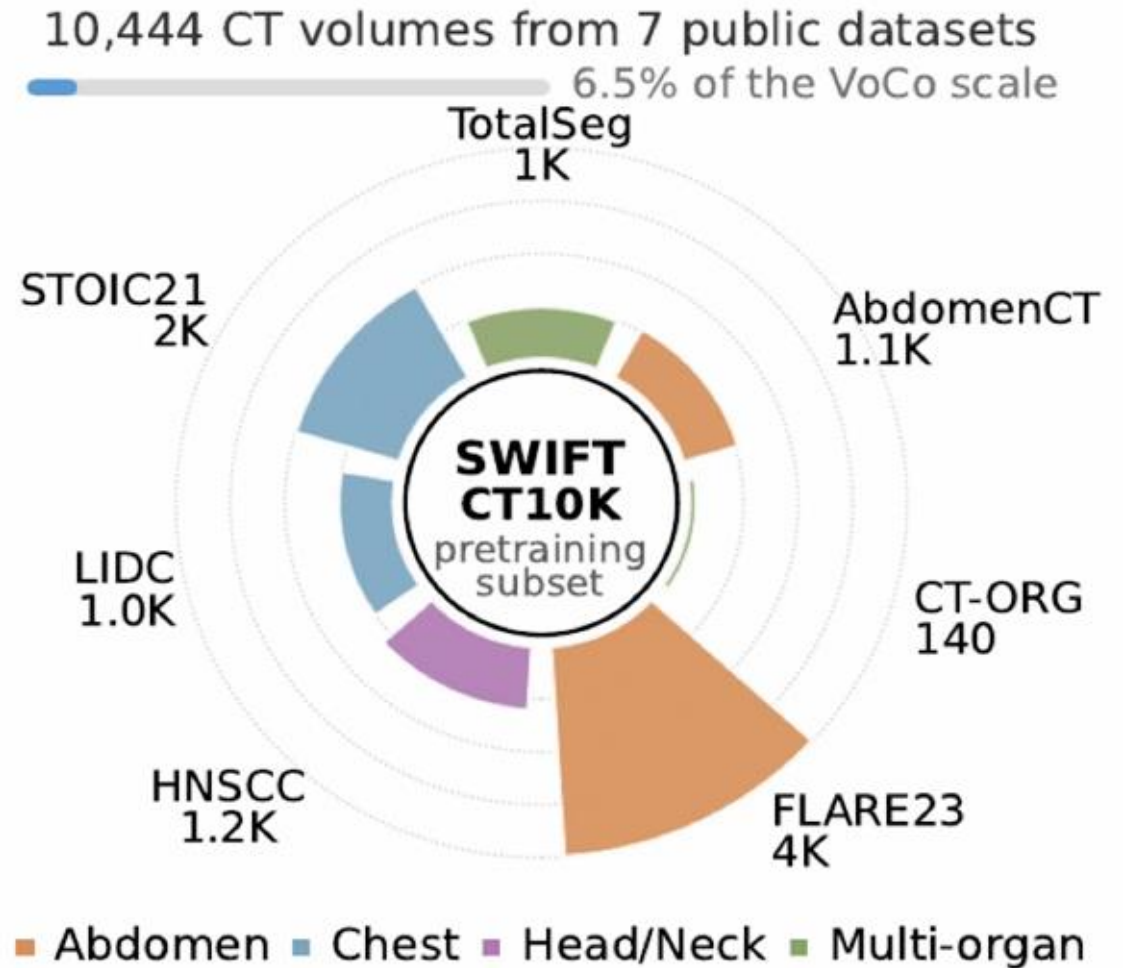


Our solution: Change pretraining approach

1. Replace Pretraining objective in SwinUNETR

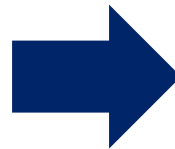
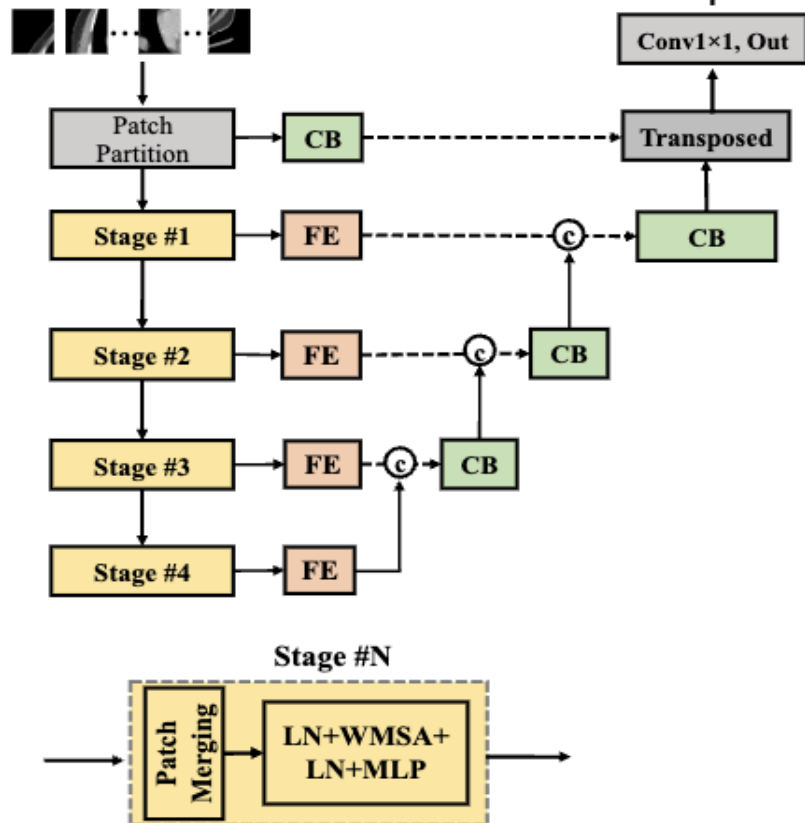
Self-distillation with masked patch prediction (DinoV2)

- ✓ Processes 3D volumes
- ✓ Replaces original Dino v2 for 3D masked patch prediction
- ✓ Pretraining used public 3D-CT datasets

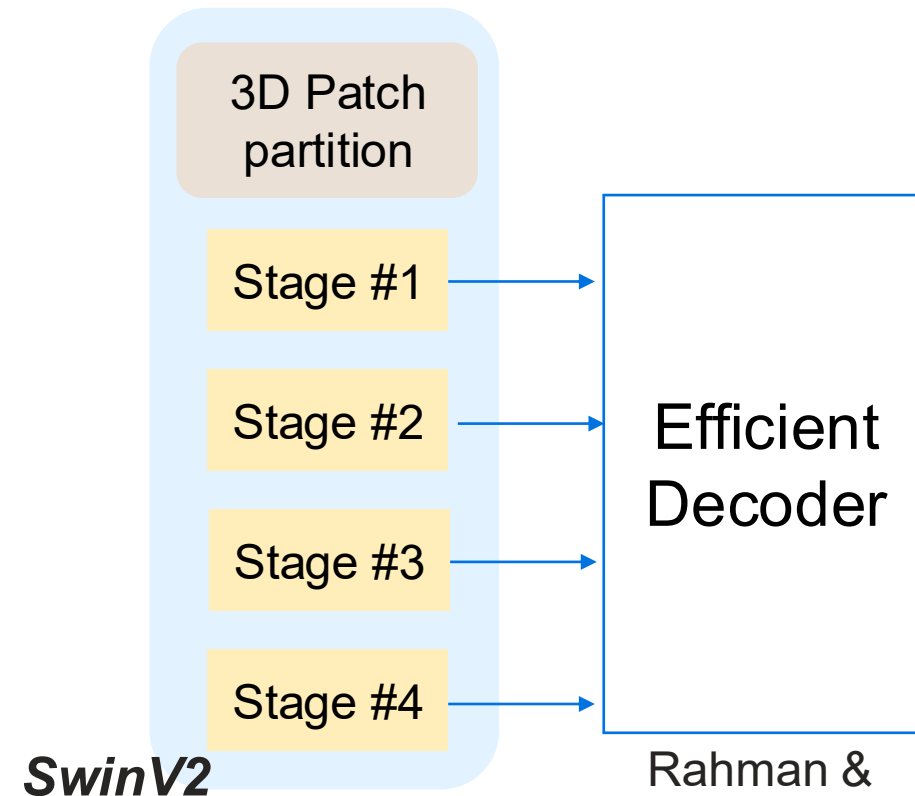
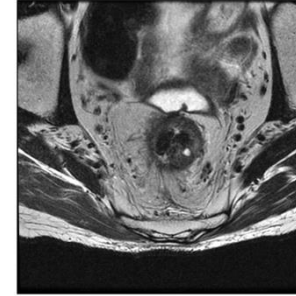


Our solution: Efficient architecture

2. Replace heavy SwinUNETR decoder with an Efficient decoder



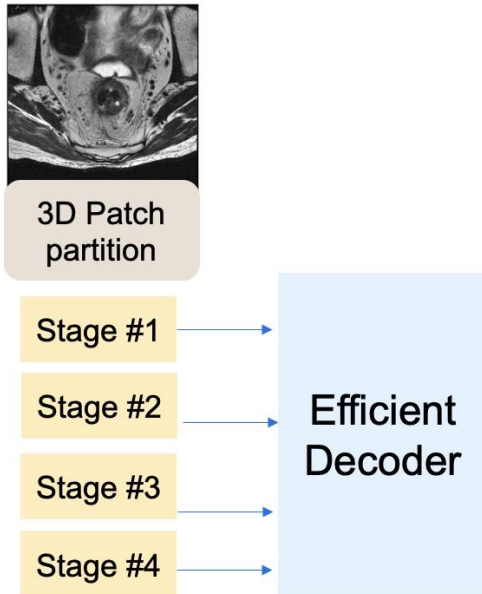
Analysis is in 3D



Rahman & Marculescu CVPR '25 ⁵

Our solution: Efficient fine tuning

2. Replace heavy SwinUNETR decoder with an Efficient decoder



3. Freeze pretrained encoder and implement Low Rank Adaptation

➤ **SWIFTe-LoRA:**

➤ LoRA adapters inserted into the Swin window query, key, value, and output projection layers

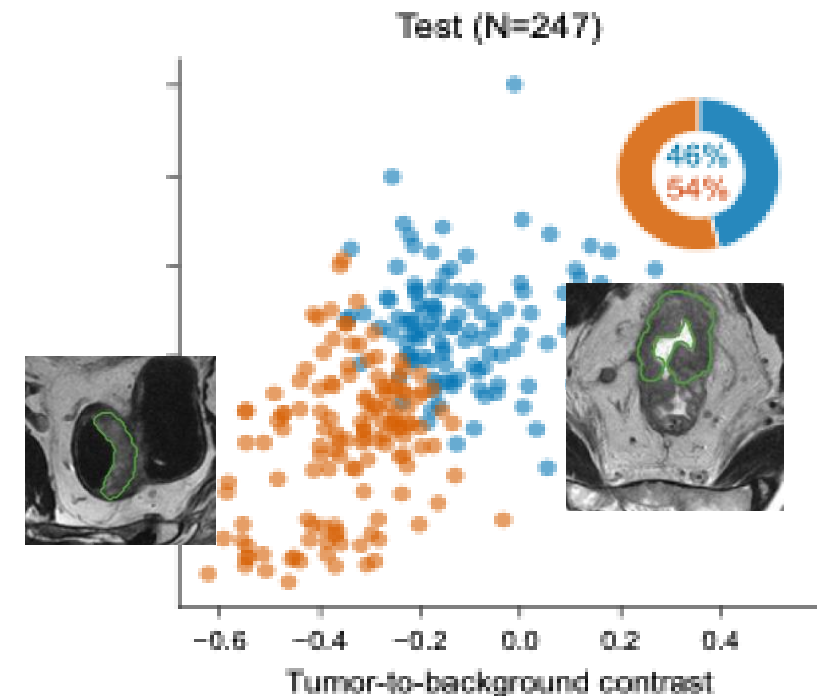
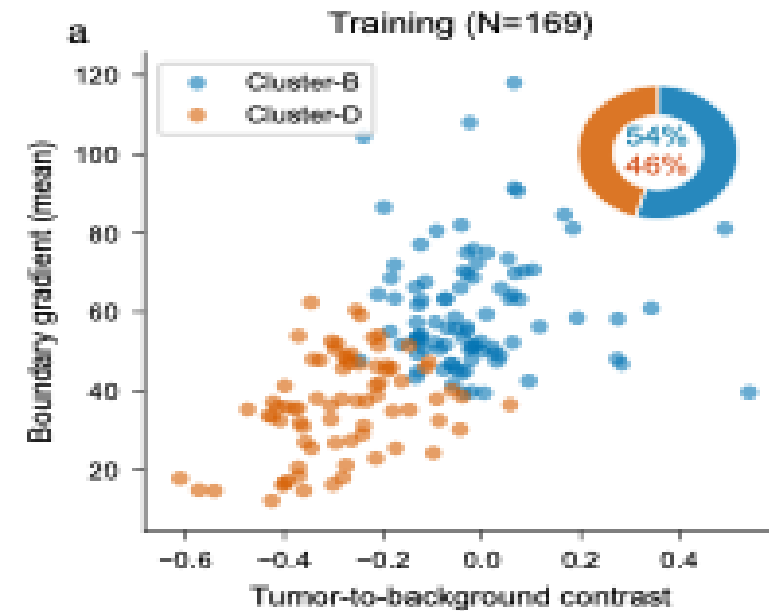
➤ **SWIFTe-LDE4:**

➤ Ensemble predictor with 4 member LoRA adapters

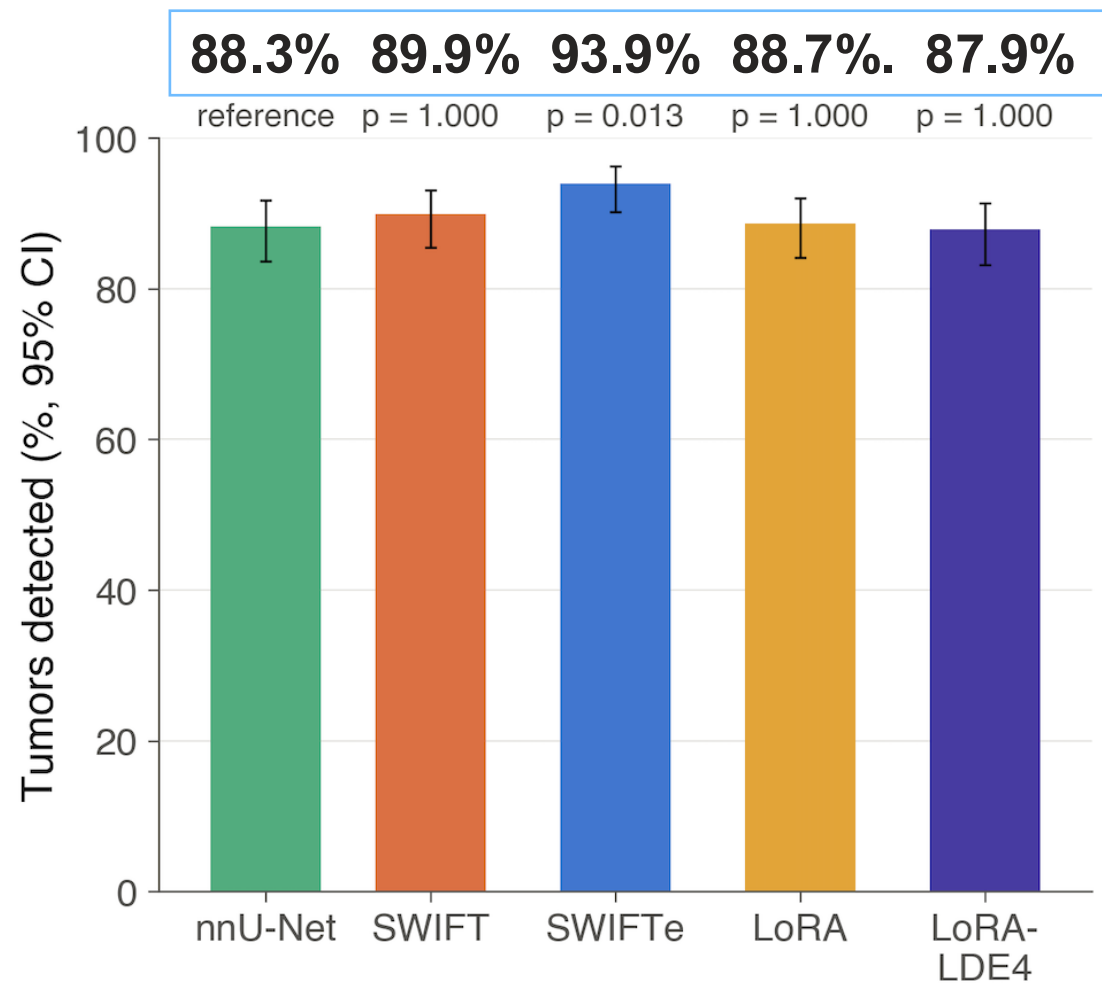
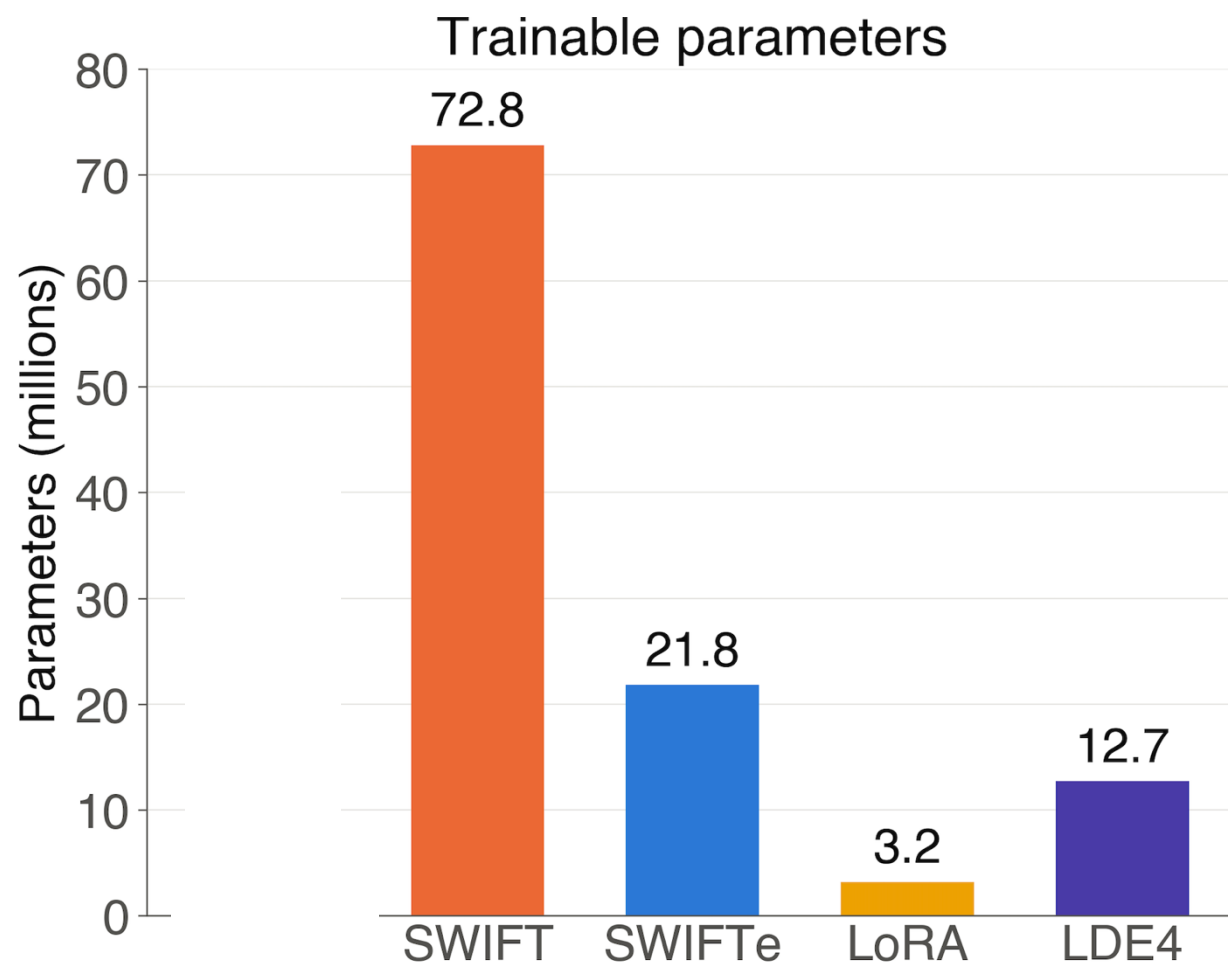
Datasets

Oblique axial T2-weighted MRI of patients with biopsy proven locally advanced rectal adenocarcinoma before total neoadjuvant treatment

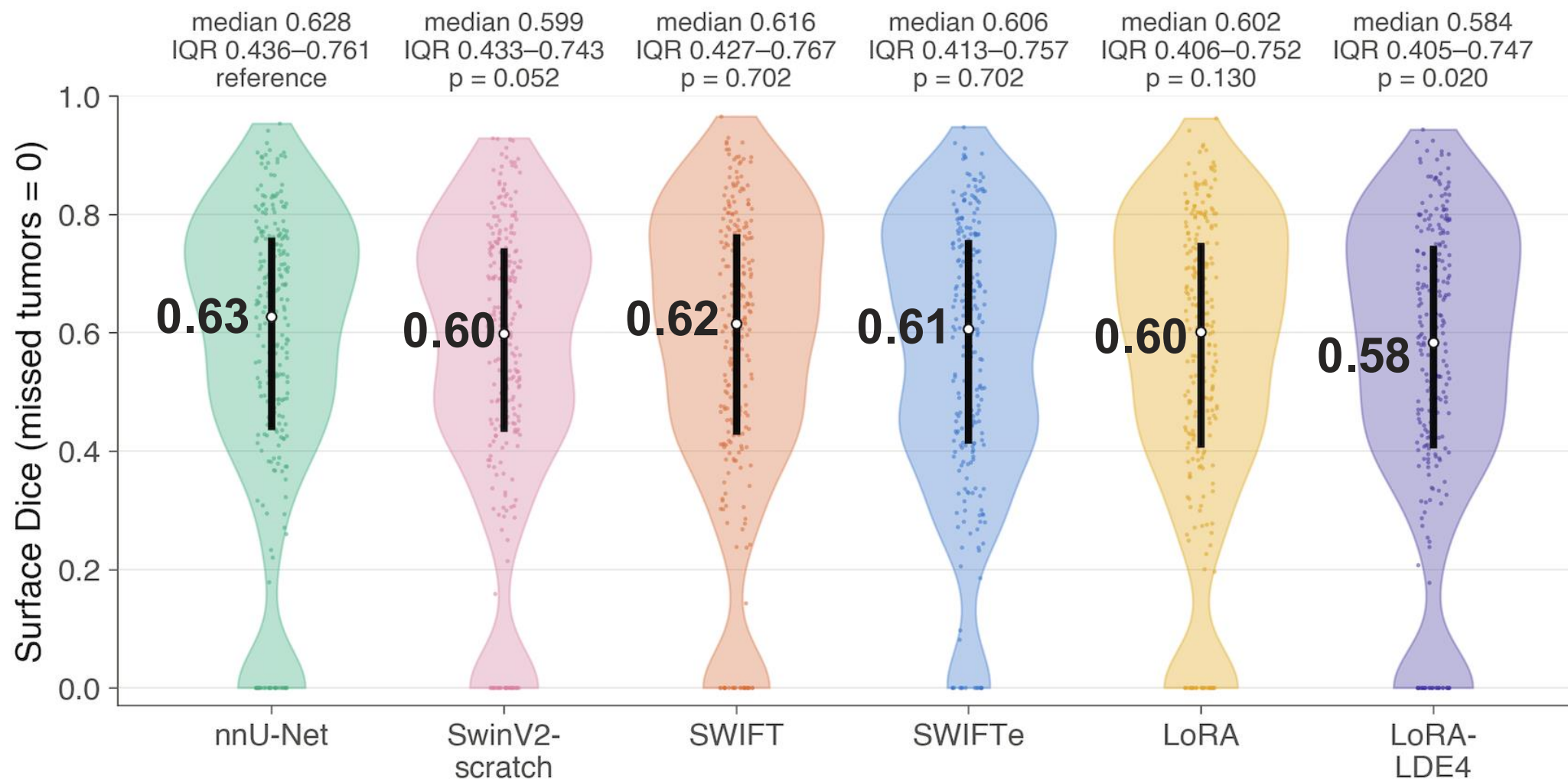
	Training	Test
N (scans)	169	247
In-plane resolution (mm)	0.43 [0.39–0.47]	0.43 [0.35–0.63]
Slice thickness (mm)	3.00–4.00	3.00–4.00
Native axial extent after resampling	88 [70–109]	93 [70–120]



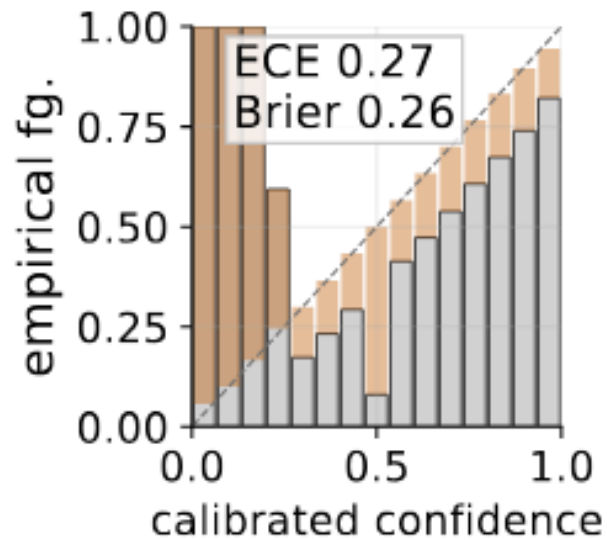
Results: Similar detection rates with fewer trainable parameters



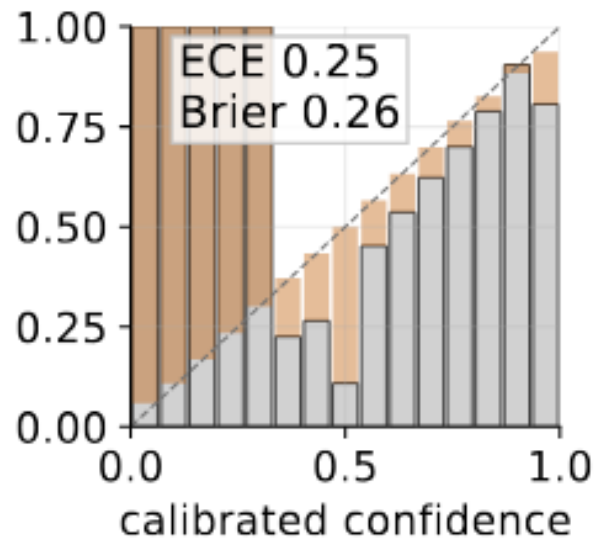
Results: Nearly similar performance across methods



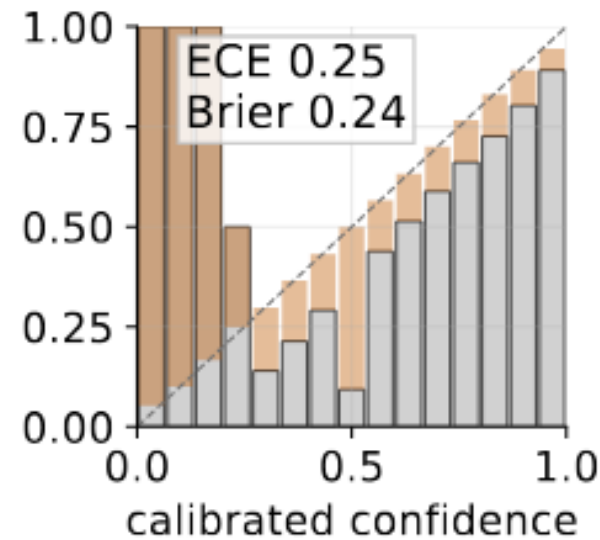
Results: Models with fewer trainable parameters were better calibrated



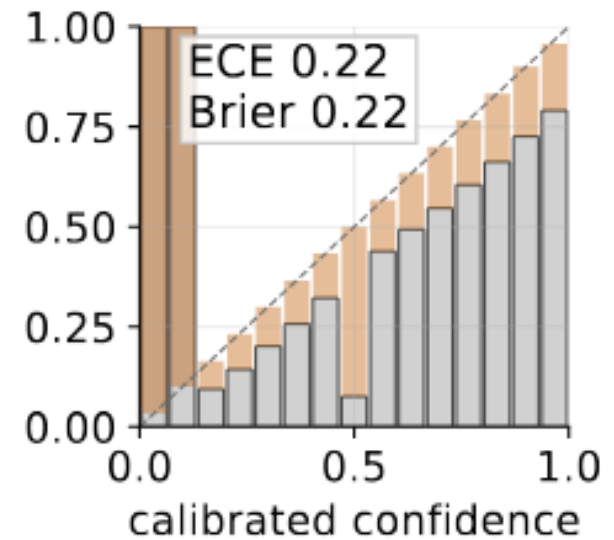
(a) SWIFT.



(b) SWIFTe.



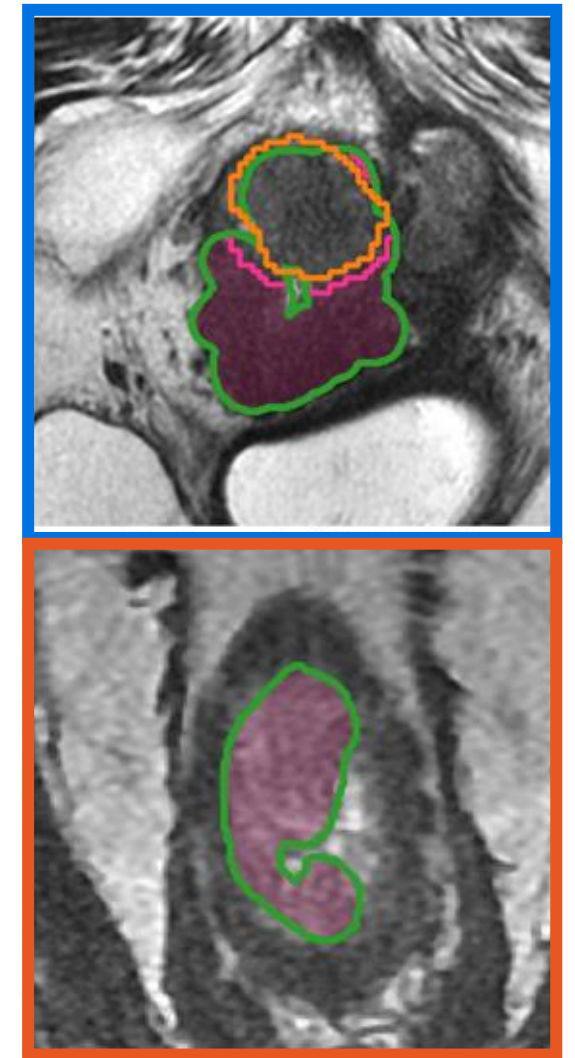
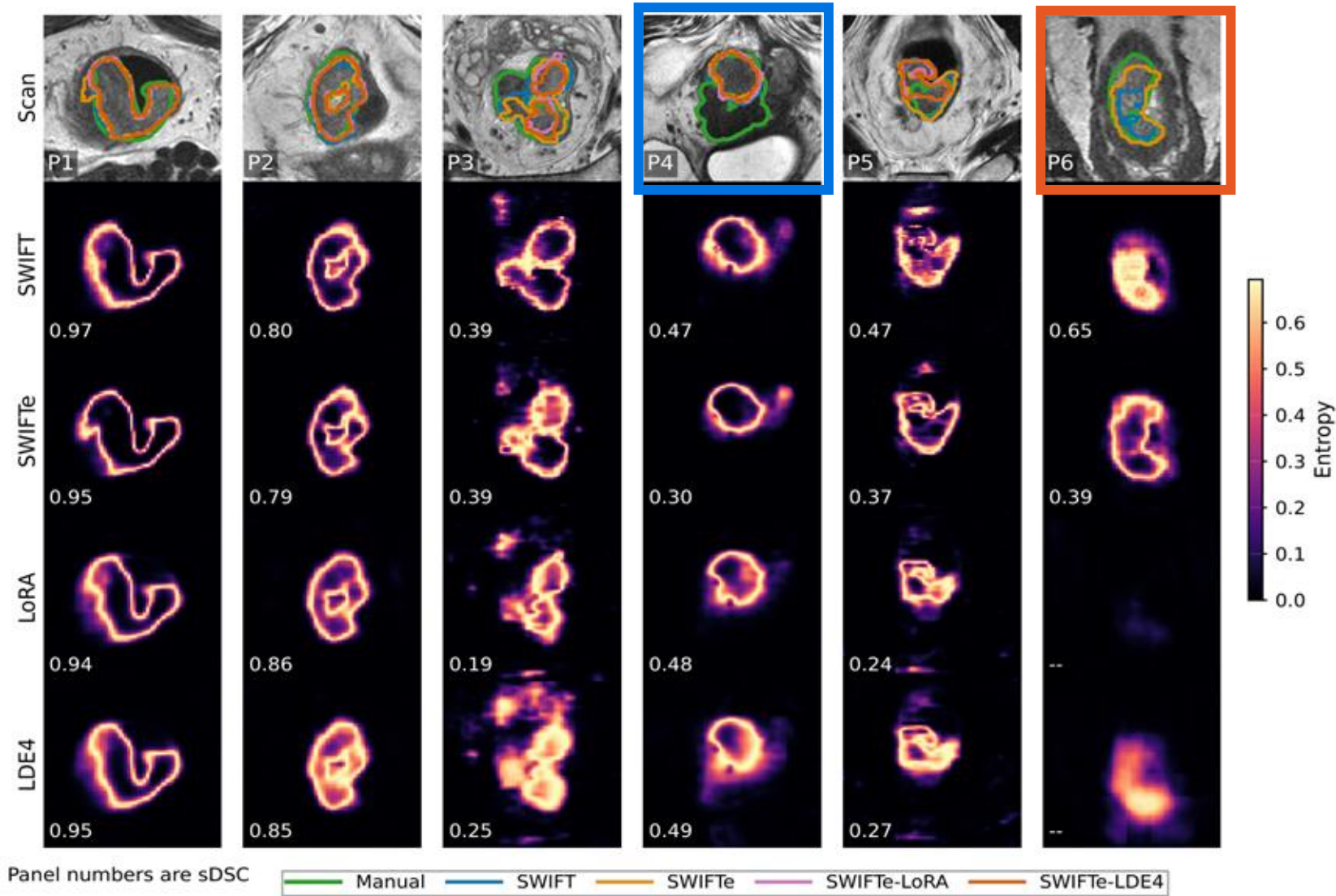
(c) SWIFTe-LoRA.



(d) SWIFTe-LDE4.

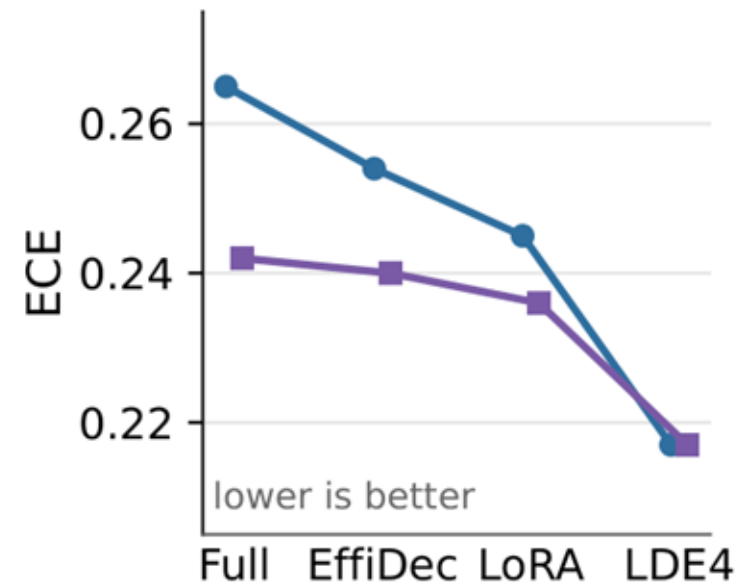
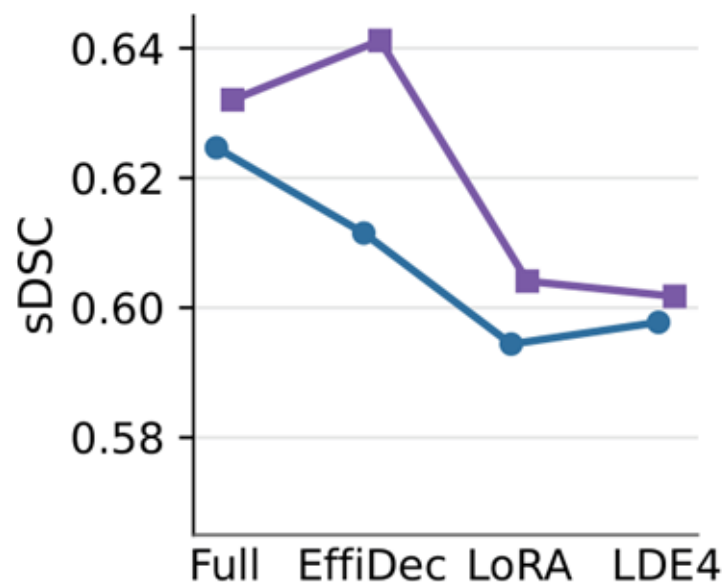
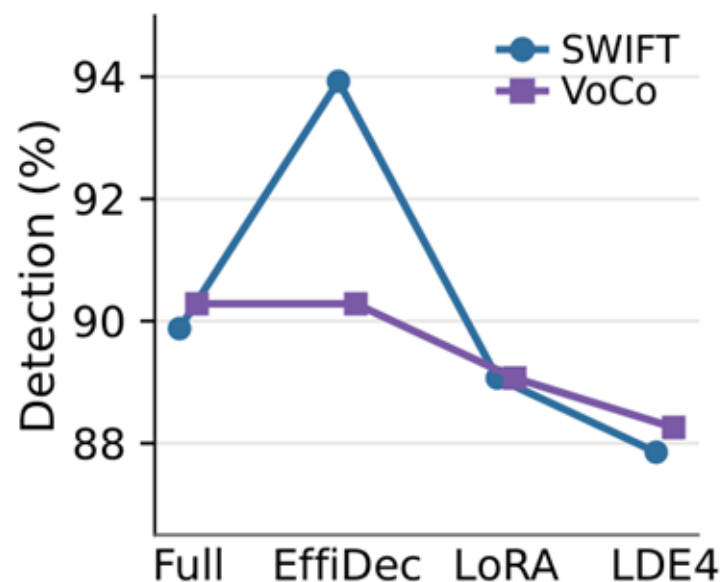
- False negatives corresponded to under-segmented tumor regions
- False negatives **decreased** with slight calibration improvement in models

Meaningful uncertainty visualization with calibration

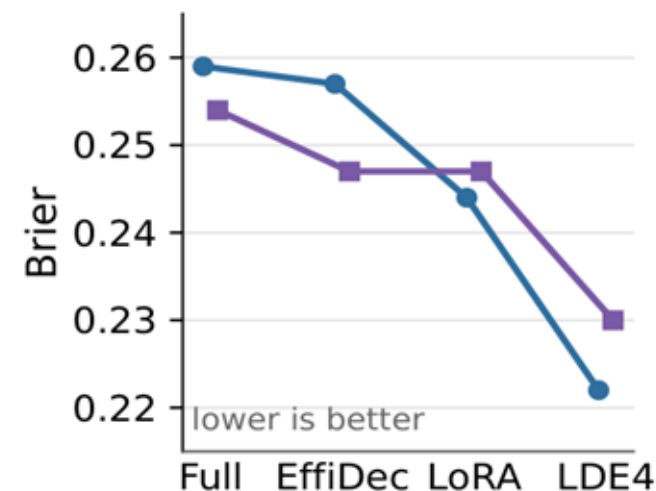


Manual tumor voxels with $p < 0.10$

Results: VoCo architecture shows similar performance



VoCo pretraining uses 106K 3D CT datasets



Conclusions and future work

- Pretraining in cross modality tasks is less beneficial especially when number of parameters that need to be fine-tuned increase
- Reducing number of trainable parameters can increase calibration without compromising performance
- Analysis on other cancer types, modalities, and cross-institutional datasets is required to establish validity



Github



Paper

Our solution: Include tumor-specific characteristics for augmentation

1. Replace Pretraining objective

2. Tumor-aware augmentation to handle tumor appearance variability

$$I'(x) = \begin{cases} \alpha \cdot I(x) + \beta & \text{if } M(x) > 0 \\ I(x) & \text{otherwise.} \end{cases}$$

Randomly samples

$$\alpha \sim N(0.5, 1.5); \beta \sim N(-0.2, 0.2)$$

